ME403 – Introduction to Robotics

Lab 00: Introduction Week Handout

Sabanci University | Spring 2025-26

1. Overview

This lab session focuses on setting up your development environment and verifying the connection with the Dobot Magician robotic arm. You will learn how to interface with the robot using Python and perform basic joint-space control.

Included Scripts

Script	Purpose and Usage
`01_init_check.py`	**Purpose:** Verifies the hardware connection, clears active alarms, and moves the robot to a safe ready position. **Usage:** Run this first to ensure your setup is functional.
`02_joint_control.py`	**Purpose:** Provides an interactive terminal interface (REPL) to move individual joints. **Usage:** Enter joint angles in degrees to observe the robot's motion.

2. Quick Start Guide

Option A: Mamba (Recommended)

```
# 1. Activate your environment
mamba activate dobot

# 2. Install requirements
pip install -r requirements.txt

# 3. Verify connection
python 01_init_check.py
```

Option B: Python venv

```
# 1. Create and activate environment
python3 -m venv .venv
source .venv/bin/activate     # Linux/macOS
# .venv\Scripts\activate      # Windows
```

```
# 2. Install requirements and verify
pip install -r requirements.txt
python 01_init_check.py
```

3. Hardware Configuration

3.1. Physical Setup Protocol

1. **Surface:** Place the robot on a flat, stable surface. Clear a 40 cm radius around the base.
2. **Posture:** Manually set the arm to a neutral posture (both segments at ≈45°) before powering on to prevent sudden torque spikes.
3. **Power:** Connect the 12V DC adapter. **USB power alone is insufficient** for motor operation.
4. **Communication:** Connect the USB cable. The robot uses a CP210x USB-to-UART bridge.

3.2. Status Signals

- **Green LED + Single Beep:** System initialized and ready.
- **Red LED + Rapid Beeping:** Alarm state (usually a collision or limit violation). Run `01_init_check.py` to clear.
- **Yellow/Flashing LED:** Busy (homing or executing a command).

4. Software and OS Setup

4.1. Linux Setup

Users must have permission to access serial devices. Run the following once:

```
sudo usermod -a -G dialout $USER
# Log out and log back in for changes to take effect.
```

The robot typically appears as `/dev/ttyUSB0` or `/dev/ttyACM0`.

4.2. Windows Setup

The driver is usually handled by Windows Update. To find the port:

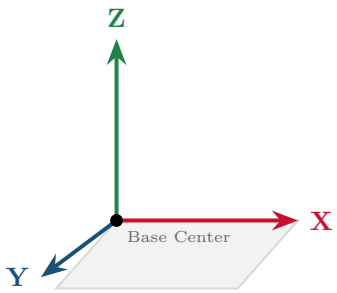
1. Open **Device Manager**.
2. Expand **Ports (COM & LPT)**.
3. Look for **Silicon Labs CP210x USB to UART Bridge** (e.g., `COM3`).

4.3. macOS Setup

Modern macOS versions include the driver. The device will appear as: `/dev/cu.usbserial-*` or `/dev/cu.SLAB_USBtoUART`.

5. Dobot Magician Reference

5.1. Coordinate System (Cartesian)



- **X-axis:** Forward (pointing away from base).
- **Y-axis:** Left/Right (Left is positive).
- **Z-axis:** Vertical (Up is positive).
- **Units:** Millimeters (mm).
- **Rotation (R):** Tool rotation in degrees.

5.2. Joint Definitions and Safe Limits

Joint	Motion	Safe Range	Units
J1	Base Rotation	±90°	deg
J2	Rear Arm Elevation	0° to 85°	deg
J3	Forearm Elevation	-10° to 85°	deg
J4	End-Effector Rotation	±90°	deg

Cartesian Dimension	Safe Operating Range	Unit
X (Reach)	120 to 315	mm
Y (Slew)	±158	mm
Z (Height)	5 to 155	mm

6. Troubleshooting

Common Connection Issues

Symptom	Recommended Action
`Access Denied` or `Port Busy`	Ensure **DobotStudio** and all other Python scripts are closed. Only one process can own the serial port.
`Device not found`	Verify both USB and Power cables. Check Device Manager (Win) or `ls /dev/tty*` (Unix).
Motors don't move	Check the 12V power supply. If the light is red, run `01_init_check.py` to clear alarms.
Erratic movement	Reset to neutral posture and power cycle the robot.

7. Safety Mandates

Mandatory Safety Rules

1. **Workspace:** Maintain a clean area. No cables or personal items in the reach zone.
2. **Initialization:** Always run `01_init_check.py` before starting your own scripts.
3. **Helper Functions:** Use `safe_move()` from `utils.py`. It enforces software limits to prevent physical damage.
4. **Supervision:** Stay within reach of the power cable for immediate emergency disconnection.